

Arithmetic Configuration Factorization

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Abstract

For a finite set of integral shifts, the residue classes on which every translate is invertible factor exactly over coprime squarefree moduli. The associated normalized local densities therefore form a positive convergent Euler product whenever the set of shifts is admissible. Convergence is automatic: outside finitely many primes, the local factor differs from one by $O_k(p^{-2})$. The pair of shifts $\{0, 2\}$ exhibits both the exceptional prime and the resulting product. Admissibility also supplies a residue class on which a divisor-incidence map gives a generalized Rayleigh quotient. A value above an integer m forces at least $m + 1$ prime translates in the sampled interval. Neither the residue class nor the local product forces the quotient above a threshold.

1 The finite product map

For fixed integer shifts, which residue classes make every translate coprime to a given modulus? Coprimality to a product of coprime moduli is equivalent to coprimality to each factor. The Chinese remainder theorem then separates the allowed classes into a product of finite sets.

Fix a nonempty finite set of distinct integers $\mathcal{H} = \{h_1, \dots, h_k\}$. For a positive integer q , set

$$R_q = \mathbb{Z}/q\mathbb{Z}, \quad G_q(\mathcal{H}) = \{a \in R_q : a + h_i \in R_q^\times \text{ for } 1 \leq i \leq k\}.$$

If q_1 and q_2 are coprime, the Chinese-remainder ring isomorphism is

$$c_{q_1, q_2} : R_{q_1 q_2} \longrightarrow R_{q_1} \times R_{q_2}, \quad [a]_{q_1 q_2} \longmapsto ([a]_{q_1}, [a]_{q_2}).$$

Proposition 1.1 (Coprime configuration factorization). *For coprime positive integers q_1, q_2 , restriction of c_{q_1, q_2} gives the bijection*

$$\gamma_{q_1, q_2} : G_{q_1 q_2}(\mathcal{H}) \xrightarrow{\sim} G_{q_1}(\mathcal{H}) \times G_{q_2}(\mathcal{H}).$$

For three pairwise coprime moduli, the two iterated product maps agree after the canonical identification of the two parenthesizations.

Proof. An element $a + h_i$ is a unit in $R_{q_1 q_2}$ if and only if its two images are units in R_{q_1} and R_{q_2} . The ring isomorphism therefore carries the defining conditions for $G_{q_1 q_2}(\mathcal{H})$ precisely to the defining conditions for the displayed product. The associativity statement follows because either iterated map sends $[a]_{q_1 q_2 q_3}$ to $([a]_{q_1}, [a]_{q_2}, [a]_{q_3})$. $\square$

For a prime p , define

$$\Omega_p(\mathcal{H}) = \{-h_i \bmod p : 1 \leq i \leq k\} \subset \mathbb{F}_p, \quad \nu_p(\mathcal{H}) = |\Omega_p(\mathcal{H})|.$$

Then $G_p(\mathcal{H}) = \mathbb{F}_p \setminus \Omega_p(\mathcal{H})$. Iterating γ gives, for every positive squarefree q , a specified bijection

$$G_q(\mathcal{H}) \xrightarrow{\sim} \prod_{p|q} (\mathbb{F}_p \setminus \Omega_p(\mathcal{H})), \quad |G_q(\mathcal{H})| = \prod_{p|q} (p - \nu_p(\mathcal{H})).$$

2 The infinite product

For a prime p , the proportion of allowed residue classes is $1 - \nu_p(\mathcal{H})/p$. Each separate condition $a + h_i \not\equiv 0 \pmod{p}$ holds on a proportion $1 - 1/p$ of the classes. Dividing the joint proportion by the product of these separate proportions gives the following local factor. This normalization makes no assertion that the conditions are independent.

Definition 2.1. *The set $\mathcal{H}$ is admissible if $\nu_p(\mathcal{H}) < p$ for every prime p . Its local factor is*

$$\sigma_p(\mathcal{H}) = \frac{1 - \nu_p(\mathcal{H})/p}{(1 - 1/p)^k}.$$

Admissibility is exactly the positivity of every local factor. It removes all congruence obstructions, but it asserts no distribution theorem.

Theorem 2.2 (Positive singular product). *If $\mathcal{H}$ is admissible, then*

$$\mathfrak{S}(\mathcal{H}) = \prod_p \sigma_p(\mathcal{H})$$

converges to a finite, strictly positive real number. More precisely,

$$\sigma_p(\mathcal{H}) = 1 - \frac{k(k-1)}{2p^2} + O_k(p^{-3})$$

outside a finite set of primes.

Proof. Put

$$\Delta_{\mathcal{H}} = \prod_{1 \leq i < j \leq k} |h_i - h_j|,$$

with the empty product equal to 1. If $p \nmid \Delta_{\mathcal{H}}$, the reductions of the h_i are distinct, so $\nu_p(\mathcal{H}) = k$. Thus, for all but finitely many primes,

$$\sigma_p = (1 - k/p)(1 - 1/p)^{-k}.$$

For $p > k$, the convergent power series for the logarithm gives

$$\log \sigma_p = - \sum_{n \geq 2} \frac{k^n - k}{np^n} = - \frac{k(k-1)}{2p^2} + O_k(p^{-3}).$$

Exponentiation gives the asserted expansion and, in particular, $|\sigma_p - 1| = O_k(p^{-2})$. Hence $\sum_p |\sigma_p - 1| < \infty$. Eventually $|\sigma_p - 1| \leq 1/2$, and then $|\log \sigma_p| \leq 2|\sigma_p - 1|$. The logarithmic series therefore converges absolutely. All its factors are positive by admissibility, so the product is finite and strictly positive. $\square$

Example 2.3 (The shifts $\{0, 2\}$). *Let $\mathcal{H} = \{0, 2\}$. At the exceptional prime 2, the two forbidden classes coincide:*

$$\Omega_2(\mathcal{H}) = \{0\}, \quad G_2(\mathcal{H}) = \{1\}, \quad \sigma_2 = \frac{1 - 1/2}{(1 - 1/2)^2} = 2.$$

For every odd prime p , the forbidden classes 0 and -2 are distinct, so

$$|G_p(\mathcal{H})| = p - 2, \quad \sigma_p = \frac{1 - 2/p}{(1 - 1/p)^2} = \frac{p(p-2)}{(p-1)^2} = 1 - \frac{1}{(p-1)^2}.$$

Consequently

$$\mathfrak{S}(\{0, 2\}) = 2 \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) > 0.$$

The positivity and convergence of this constant are local statements. The assertion that infinitely many integers n make both n and $n+2$ prime is not a consequence of them.

3 A finite divisor-sum criterion

To detect several prime translates at one integer, one can compare a nonnegative weighted prime count with the total weight. If the weighted average exceeds an integer m , some integer with positive weight has more than m prime translates. Squared divisor sums provide nonnegative weights whose total weight can be computed from congruences.

Let $\mathcal{H}$ be admissible, and let W be a positive squarefree integer divisible by every prime factor of $\Delta_{\mathcal{H}}$. The finite product map gives

$$|G_W(\mathcal{H})| = \prod_{p|W} (p - \nu_p(\mathcal{H})) > 0.$$

Choose an integer $v_0 = v_0(W)$ whose class lies in $G_W(\mathcal{H})$. Thus $\gcd(v_0 + h_i, W) = 1$ for every i . Let N be a positive integer such that

$$I_N = \{n \in \mathbb{Z} : N \leq n < 2N, n \equiv v_0 \pmod{W}\}$$

is nonempty and $n + h_i \geq 2$ for $n \in I_N$ and all i . Let $\mathcal{D}$ be a finite set of k -tuples of positive integers containing $\mathbf{1} = (1, \dots, 1)$. Require

$$d_1 \cdots d_k \text{ to be squarefree,} \quad \gcd(d_1 \cdots d_k, W) = 1 \quad (\mathbf{d} \in \mathcal{D}).$$

Define the divisor-incidence map by

$$T : \mathbb{R}^{\mathcal{D}} \longrightarrow \mathbb{R}^{I_N}, \quad (T\lambda)(n) = \sum_{\substack{\mathbf{d} \in \mathcal{D} \\ d_i | n + h_i \ (1 \leq i \leq k)}} \lambda_{\mathbf{d}}$$

and assign to $n \in I_N$ the weight

$$w_{\lambda}(n) = |(T\lambda)(n)|^2 \geq 0.$$

These squared divisor sums are finite multidimensional Selberg weights as in [1, (2.4), (4.2)–(4.3)]. To count prime translates, put

$$R_{\mathcal{H}}(n) = \sum_{i=1}^k \mathbf{1}_{\mathbb{P}}(n + h_i), \quad (M_R f)(n) = R_{\mathcal{H}}(n) f(n),$$

where $\mathbb{P}$ is the set of positive primes, and define

$$A = T^* T, \quad B = T^* M_R T,$$

with respect to the standard Euclidean inner products.

Proposition 3.1 (Finite divisor-sum Rayleigh principle). *For $\mathbf{d}, \mathbf{e} \in \mathcal{D}$, set $q_i = \text{lcm}(d_i, e_i)$. The matrix entries are*

$$A_{\mathbf{d}, \mathbf{e}} = \#\{n \in I_N : q_i \mid n + h_i \text{ for } 1 \leq i \leq k\}$$

and, when the q_i are pairwise coprime,

$$\left| A_{\mathbf{d}, \mathbf{e}} - \frac{N}{W \prod_{i=1}^k q_i} \right| \leq 1.$$

If they are not pairwise coprime, then $A_{\mathbf{d}, \mathbf{e}} = 0$. Moreover,

$$B_{\mathbf{d}, \mathbf{e}} = \sum_{\substack{n \in I_N \\ q_i \mid n + h_i \ (1 \leq i \leq k)}} R_{\mathcal{H}}(n).$$

For $\lambda \in \mathbb{R}^{\mathcal{D}}$, write

$$S_1(\lambda) = \|T\lambda\|_2^2, \quad S_2(\lambda) = \langle T\lambda, M_R T\lambda \rangle.$$

If $V = (\ker T)^\perp$ and A_V, B_V denote the restrictions to V , then A_V is positive definite and

$$\max_{\lambda \notin \ker T} \frac{S_2(\lambda)}{S_1(\lambda)} = \lambda_{\max}(A_V^{-1/2} B_V A_V^{-1/2}) \leq \max_{n \in I_N} R_{\mathcal{H}}(n) \leq k.$$

Consequently, if m is a nonnegative integer and some λ satisfies

$$S_1(\lambda) > 0, \quad S_2(\lambda) > m S_1(\lambda),$$

then at least $m + 1$ of $n + h_1, \dots, n + h_k$ are prime for some $n \in I_N$.

Proof. The conditions on $\mathcal{D}$ imply that $W, q_1, \dots, q_k$ are individually squarefree and that $\gcd(W, q_i) = 1$. If a prime p divides both q_i and q_j with $i \neq j$, simultaneous divisibility would give $h_i \equiv h_j \pmod{p}$. But then $p \mid \Delta_{\mathcal{H}}$, hence $p \mid W$, contrary to $\gcd(W, q_i q_j) = 1$. Thus the congruences counted by $A_{\mathbf{d}, \mathbf{e}}$ are inconsistent unless the q_i are pairwise coprime. In the remaining case, the Chinese remainder theorem gives one residue class modulo $W \prod_i q_i$. Its intersection with an interval of N consecutive integers has cardinality differing from $N/(W \prod_i q_i)$ by at most 1. Expanding the two inner products gives the displayed entries of A and B .

Since $\mathbf{1} \in \mathcal{D}$ and I_N is nonempty, T is nonzero. If $0 \neq x \in V$, then

$$\langle x, A_V x \rangle = \|Tx\|_2^2 > 0,$$

so A_V is positive definite. Both A and B annihilate $\ker T$ and preserve V . Orthogonal projection onto V , followed by the substitution $y = A_V^{1/2} x$, therefore reduces the generalized quotient to

$$\frac{\langle x, B_V x \rangle}{\langle x, A_V x \rangle} = \frac{\langle y, A_V^{-1/2} B_V A_V^{-1/2} y \rangle}{\langle y, y \rangle}.$$

Rayleigh–Ritz gives the equality with the largest eigenvalue. Finally,

$$S_2(\lambda) = \sum_{n \in I_N} R_{\mathcal{H}}(n) |(T\lambda)(n)|^2 \leq \left(\max_{n \in I_N} R_{\mathcal{H}}(n) \right) S_1(\lambda).$$

If $S_2(\lambda) > m S_1(\lambda)$, some term with $(T\lambda)(n) \neq 0$ has $R_{\mathcal{H}}(n) > m$. This integer-valued prime count is therefore at least $m + 1$. $\square$

Example 3.2 (Two divisor coefficients). Take $\mathcal{H} = \{0, 2\}$, $W = 2$, $v_0 = 1$, $N = 5$, and $\mathcal{D} = \{(1, 1), (3, 1)\}$. Then $I_N = \{5, 7, 9\}$. The first divisor tuple contributes at every sample. The second contributes only at $n = 9$, since $3 \mid 9$ but $3 \nmid 5, 7$. With rows ordered as 5, 7, 9 and columns as (1, 1), (3, 1), this gives

$$T = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad (R_{\mathcal{H}}(5), R_{\mathcal{H}}(7), R_{\mathcal{H}}(9)) = (2, 1, 1).$$

Indeed, the pairs of translates are (5, 7), (7, 9), and (9, 11). Thus

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 1 \\ 1 & 1 \end{pmatrix}.$$

For $\lambda = (a, b)$, the weights are $a^2, a^2, (a + b)^2$, so

$$\frac{S_2(\lambda)}{S_1(\lambda)} = \frac{3a^2 + (a + b)^2}{2a^2 + (a + b)^2} \leq \frac{3}{2} \quad (\lambda \neq 0).$$

Equality holds at $(a, b) = (1, -1)$, where $S_1 = 2$ and $S_2 = 3$. The strict threshold with $m = 1$ therefore detects a pair of prime translates in this interval, realized at $n = 5$. The maximum $3/2$ is smaller than the largest sampled prime count, which is 2. Every permitted weight gives equal mass to 5 and 7, so these weights cannot isolate the sample 5. This finite calculation supplies no asymptotic prime-count estimate.

Admissibility supplies the residue class $v_0 \pmod{W}$, but it supplies no strict inequality between S_2 and mS_1 . Neither positivity of $\mathfrak{S}(\mathcal{H})$ nor the finite identities above implies the threshold inequality. No limiting operator or asymptotic estimate is asserted here.

References

- [1] J. Maynard, *Small gaps between primes*, Ann. of Math. (2) **181** (2015), no. 1, 383–413, doi:10.4007/annals.2015.181.1.7.